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**Title: Single Pass Diafiltration Integrated into a Fully Continuous mAb  
Purification Process<sup>†</sup>**

**Authors and Affiliations:** Joanna Rucker-Pezzini, Lindsay Arnold, Kevin Hill-Byrne, Tom Sharp, Maksim Avazhanskiy, Chris Forespring  
MedImmune, Gaithersburg, MD 20878

**Corresponding Author:** Joanna Rucker-Pezzini, One MedImmune Way, Gaithersburg MD 20878; (301)398-1404; pezzinij@medimmune.com

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## Abstract

The concept of continuous manufacturing has gained significant interest from the biopharmaceutical industry over the past several years. Benefits include increased manufacturing productivity, improved quality control, reduction in plant footprint, and more flexible management of facility capacity. There are several technologies currently available that enable continuous processing for chromatography and ultrafiltration. However, a single pass diafiltration design that meets the required small molecule clearance and has been integrated into a fully continuous monoclonal antibody purification process has not been previously published. Here, the theory and design of a 3-stage single pass diafiltration step is presented. Buffer exchange greater than 99.75% was experimentally demonstrated. Several critical design aspects were incorporated to minimize system complexity and reduce the buffer volume requirements. Lastly, single pass diafiltration was demonstrated in a pilot scale continuous process with uninterrupted flow from the bioreactor through the formulation step. This work illustrates the feasibility of incorporating a single pass diafiltration step into an end-to-end continuous protein purification process. This article is protected by copyright. All rights reserved

## Introduction

Continuous manufacturing is an established production mode for a range of industries including petrochemical and food production (Anderson, 2001; Reay et al, 2013). However, this concept has not been widely implemented in the biopharmaceutical industry. To date, only two FDA approved small molecules, and zero approved biological products, report the use of continuous manufacturing despite the opportunities (Queensbury et al., 2016). Benefits of a continuous process include consistent product quality, higher productivity, reduced equipment size, steady state operation, and streamlined process flow (Cooney and Konstantinov, 2014). The ability to control batch volumes based on both equipment size (scale-up) and batch duration (scale-out) provides significant flexibility for both large and small product demands. The improved productivity derived from continuous technology may be an ideal solution for medicines that require many batches to meet commercial demand. Alternatively, the ability to process small volumes has advantages for precision medicine applications. Additionally, as perfusion harvest methods become increasingly productive, a downstream design that can efficiently process large volumes at low titers will be crucial.

A number of efforts have been made to assemble a continuous process from bioreactor through purification (Godawat et al., 2015; Klutz et al., 2015; Levison, 2016). While some unit operations are ready for commercialization, such as multicolumn chromatography (Bisschops et al., 2009), other technology gaps remain (Zydney, 2016; Croughan et al., 2015). In this study, a continuous diafiltration step was designed and implemented into a fully continuous pilot scale production train designed to process a 200L fed-batch bioreactor (1 kilogram antibody) in four days.

Diafiltration is performed for buffer exchange during formulation of the therapeutic protein to achieve an appropriate buffer composition (Bolton et al., 2011). Currently, buffer exchange is typically performed with systems that use a recirculation loop to and from a tangential flow filtration (TFF) membrane and a feed tank large enough to hold the entire batch volume. This will be referred to as "recirculated diafiltration". The TFF membrane allows components smaller than the molecular weight cut off to pass through the filter and into the permeate stream directed to waste, while larger components are retained in the retentate stream and recirculated back to the feed tank. A required volume of diafiltration buffer, specified to attain a desired clearance factor, is added to the feed tank or retentate stream at the same rate that buffer is removed through the permeate to maintain a constant volume in the feed vessel (Schwartz, 2003). The permeate flux (flow rate / membrane area) is usually up to thirty percent that of the feed flux, and is dependent on the product concentration. Typically, a diafiltration buffer volume of at least six times the initial product volume, also stated as six diafiltration volumes or diavolumes (DVs), is used to attain 99.75% or greater buffer exchange, and meet the recommendations in ICH Guideline Q3C, Impurities: Residual Solvents.

The technology outlined in this paper enables ultrafiltration and diafiltration in a continuous purification process through the use of single pass tangential flow filtration (SPTFF) modules. SPTFF modules contain typically two or more TFF filter units in series to achieve high protein concentration factors in a single pass through the filters. This eliminates the need to recirculate product through the membrane multiple times to achieve ultrafiltration. In addition to long membrane path lengths, low feed fluxes are used to increase membrane residence time, and as a result, increase the concentration factor (Dizon-Maspat et al., 2012). Membranes can be configured with a step-wise decrease in the cross-sectional membrane area as the membranes are

placed in series within a SPTFF module, which results in higher concentration factors at a given feed flux than a design that keeps the cross-sectional membrane area constant (Casey, 2011). For the method proposed, diafiltration is achieved in multiple stages of concentration with commercially available SPTFF modules followed by dilution in an interim surge vessel (Jungbauer, 2013). This technology will be referred to as "single pass diafiltration" (SPDF) and is illustrated in Figure 1. Each stage consists of a feed pump, a SPTFF membrane and a stirred retentate surge vessel. All flow is uni-directional; retentate flow from a given stage is fed to the following stage and there is no recirculation of retentate flow back into the feed tank. Diafiltration buffer is transferred to the retentate vessel of each stage at the same rate that buffer is removed through the permeate to maintain consistent concentrations across each retentate vessel; the concentration of the SPDF feed material is the same as that of the diluted material in all retentate vessels. Feed flux is controlled to achieve the concentration factor needed for the required buffer exchange.

The theoretical final concentration ( $C_f$ ) of buffer component in the system can be calculated as shown in Equation 1, given the initial buffer component concentration ( $C_o$ ), the dilution factor ( $X$ ), and the number of dilution stages ( $n$ ). For recirculated diafiltration, the final concentration is calculated with the total number of diavolumes ( $N$ ), as shown in Equation 2 (Tutunjian, 1983). The equations listed illustrate ideal conditions and do not include a rejection coefficient (sieving coefficient is equal to 1).

$$C_f = C_o \left( \frac{1}{X} \right)^n \quad (1)$$

$$C_f = C_o e^{-N} \quad (2)$$

For a given clearance factor ( $C_o/C_f$ ), the dilution factor and total buffer volume required

decrease exponentially as the number of stages increases, as illustrated in Figure 2 and Figure 3. A system with a high number of stages and minimal concentration and dilution of the product, has the most efficient use of diafiltration buffer. The recirculated diafiltration system described above is more efficient than any sequential diafiltration, where one performs a sequence of concentrations and dilution steps. However, the amount of buffer needed to dilute a specified product mass in a sequential diafiltration design can be minimized by an increase in the number of stages or dilution of the product at the highest possible concentration, as shown in Figure 3. Permeate flux decreases as the product concentration increases due to concentration polarization and/or membrane fouling, leading to potentially longer process times for recirculated ultrafiltration diafiltration (UFDF) processes (Ng et al., 1976). The optimal concentration for recirculated diafiltration of the antibody used in this work was experimentally determined to be 70 g/L (data not shown) and requires 0.09 L<sub>buffer</sub>/g<sub>protein</sub>. A 4-stage SPDF system that can concentrate product to 160 g/L can exceed the buffer efficiency of a recirculated diafiltration system. A 3-stage SPDF system that can concentrate product upwards of 210 g/L can meet the buffer demand of recirculated diafiltration systems. A two stage SPDF system cannot meet the buffer demand of a recirculated diafiltration system within attainable product concentration ranges. Although increasing the number of stages reduces the buffer requirements, system complexity and cost greatly increase. For this reason, a 3-stage system was developed. The primary aim of this study was to determine the optimal operating conditions to meet buffer exchange specifications while minimizing buffer requirements for a 3-stage design. The final SPDF design was then demonstrated in a pilot scale continuous process with uninterrupted flow from the bioreactor through the formulation step.

## **Materials & Methods**

### **SPDF System Set Up**

Pall Corporation Cadence Inline Concentrator (ILC) 4-in-series Delta regenerated cellulose membranes, with a molecular weight cutoff of 30 kD, were used for all diafiltration studies. Modules are designed with a step-wise decrease in membrane cross sectional area. The individual membrane area of each module was 0.065 m<sup>2</sup> and the total membrane area of the UFDF system with four membranes was 0.26 m<sup>2</sup>. Masterflex pumps with size 16 tubing provided a flow rate range of 4.8 to 480 ml/min and Ashcroft pressure gauges with a range of 0 to 60 psi are also included in the SPDF system design.

### **Equipment for Continuous End-to-End Process**

The continuous end-to-end process is outlined in Table I. Table I lists the materials and equipment associated with each process step.

The following equipment was applied throughout the process. All pumps were Masterflex L/S Digital pumps. Connective tubing was Masterflex size 13 and size 16 tubing, and tubing fed through the pump heads was PharmEd tubing. Cole Parmer polypropylene hose barb to luer connections, 3-way stop cocks, and caps were used for the flow path to connect materials and equipment. VWR glass bottles with stir bars were used for break vessels and were placed on VWR stir plates. Pendotech single-use sensors and Ashcroft pressure gauges were used for pressure measurement, and Mettler Toledo pH meters and probes were used for pH measurement.

## **SPDF Feed Material**

A MedImmune recombinant monoclonal antibody (mAb) was used in all SPDF experiments. It was produced by mammalian cell culture methods and purified in batch mode according to steps listed in Table I, which are representative of a typical mAb purification process. The base buffer for the SPDF feed material and the diafiltration buffer contained histidine and arginine salts from J.T. Baker. Cupric sulfate used to spike the feed material was supplied by J.T. Baker.

## **Analytical Methods**

Protein concentrations were measured with a C Technologies Solo VPE at 280nm. Inductively coupled plasma-mass spectrometry (ICP-MS) analysis was used to measure quantitative trace elements of in-process samples in the range of parts per billion (ppb). Samples are dissolved in nitric acid and introduced into the ICP-MS system where they are nebulized. The aerosols are swept into the plasma torch by argon gas. In the plasma the aerosols are desolvated and ionized into positive singly charged ions and detected in the mass analyzer by their isotopic mass/charge ( $m/z$ ) ratio. The amount of copper ( $^{63}\text{Cu}$ , Helium Mode) is calculated based on the intensities of the measured samples to those obtained from standards. The analytical assay utilized Agilent 7900ce ICP-MS with auto-sampler to accurately quantitate the amount of soluble copper, as well as other trace elements in in-process samples.

## **Design of Experiments Study to Evaluate Concentration Factor**

A DOE study was performed on a single SPTFF membrane to determine the range of operating conditions that can be used to achieve the 7.4 VCF or greater required for a 3-stage



diafiltration step with a minimum of 99.75% buffer exchange. The VCF was defined as the ratio of the feed flow rate to the retentate flow rate. The experimental conditions for the DOE study are listed in Table II. The experimental variables were feed concentration, feed flux, and feed pressure. The ranges tested were determined from initial scouting studies not described here. The feed flux and pressure ranges ensured that product did not reach concentrations that fouled the membranes and ensured that the concentration factors required to achieve the diafiltration goals were observed. Feed pressure to the Pall ILC was measured rather than transmembrane pressure (TMP) or retentate pressure due to the internal restrictor on the retentate side of the SPTFF module.

Prior to use, the membranes were flushed with a minimum of 15 L/m<sup>2</sup> of 0.1N NaOH and held for >8 hours. Then, membranes were flushed with 15 L/m<sup>2</sup> reverse osmosis deionized water (RODI) and equilibrated with a minimum of 15 L/m<sup>2</sup> of diafiltration buffer. After operation, the membranes were flushed with a minimum of 15 L/m<sup>2</sup> RODI, followed by 15 L/m<sup>2</sup> of 0.1N NaOH, and stored in 0.1N NaOH. Feed flow rate was controlled by a Masterflex peristaltic pump with a digital flow set point. Calibration of the pump was performed prior to each experiment. Once the feed flow rate was established, a target feed pressure was reached by adjustment of a manual pinch valve on the retentate line. Pressure was monitored with an Ashcroft pressure gauge. For each experiment, the system was allowed to stabilize for 15 minutes before the product concentration was measured offline. Flux and pressure experiments were performed in a randomized order to prevent bias in the results.

## **Diafiltration Study**

A diafiltration study was performed to demonstrate that the expected buffer exchange could be achieved. For this study, antibody product was spiked with cupric sulfate and run through the 3-stage SPDF system. The blue coloration of cupric sulfate provided both a visual and measured change in concentration of this buffer component. The cupric sulfate concentration was measured with ICP-MS, as described in the Analytical Methods section.

The diafiltration equipment setup consists of three concentration and dilution stages setup in series, as shown in Figure 1. Each stage consists of a feed pump, a pressure gauge, an SPTFF membrane and a stirred retentate vessel. The product in the retentate vessel of each stage is the feed material for the subsequent stage. Diafiltration buffer was added to each retentate vessel to dilute the product and maintain the target feed concentration. Feed flow rates were controlled by the feed pump to achieve the desired concentration factor. No pressure control was applied.

The number of pumps for the 3-stage diafiltration setup was minimized by use of a multi-channel peristaltic pump head to supply feed flow. To maintain constant retentate vessel volumes for the duration of the run and to minimize the number of pumps required for the process, diafiltration buffer from a single buffer vessel was siphoned into the three retentate vessels, as shown in Figure 1. All retentate vessels were placed on a single stir plate and used top mounted stir bars to ensure adequate mixing. The 3-stage diafiltration equipment occupies the same bench space as the recirculated diafiltration equipment to process an equivalent batch size.

## **Continuous End-to-End Process**

Single pass diafiltration was demonstrated in an end-to-end pilot scale continuous process. The design of the fully continuous train is based on a platform process used to purify monoclonal

antibodies. As shown in Table I, the process had six steps with continuous flow from one to the next. Each step downstream of the harvest is separated by a small vessel, with continuous flow in and out, to achieve necessary pressure breaks in the flow path.

Alternating tangential flow filtration (ATF) was used to harvest the product from a bioreactor with 30 L working volume that operated in perfusion mode at 1.5 vessel volumes per day (vvd). A multicolumn chromatography system was implemented for the protein A purification step. This was followed by a low pH viral inactivation step, in which the product was inline titrated to a pH between 3.4 and 3.6, held for a specified time to achieve the required log reduction value (LRV) of virus and then neutralized to a pH between 7.4 and 7.6. A hold time with continuous flow was achieved by using a column packed with size exclusion (SEC) resin, sized to provide a residence time between 15 and 60 minutes (or as required based on viral clearance studies). The viral inactivation step preceded a filter train that contained a depth filter, a SPTFF unit to concentrate the product, a sterile filter, an anion (AEX) membrane, and a viral filter. A Millipore SPTFF unit that contained two membranes in series without a pressure break was used at this step to provide the approximate 2-fold concentration factor required to meet the SPDF feed concentration target of 15 g/L; the Pall 4-in-series SPTFF units used for the diafiltration work would result in an over-concentration of the product at the flow rates required. The last step of the continuous end-to-end process included SPDF and final concentration also with a Pall SPTFF module.

The bioreactor and Protein A step operated continuously for six days to accumulate sufficient product prior to initiation of the downstream purification process, which ran in conjunction with the upstream steps for six hours. Throughput differences between the upstream and downstream portions of the run were due to equipment availability at the time. Two six hour

purification runs were performed in which continuous flow was achieved from the bioreactor through to SPDF and final concentration. Data is shown for the second continuous run. A SPDF feed flux of 12 LMH ( $\text{L}/\text{m}^2/\text{hr}$ ) processed 11.7 g mAb per hour and determined the flow rate of the entire downstream purification train. This rate of operation allows for the purification of a “pilot scale” 200 L fed-batch bioreactor (1 kilogram mAb) in four days.

## Results

### DOE for SPDF Parameter Optimization

An initial DOE study determined the feed concentrations, feed flux and feed pressures to be used in the SPDF study. The goal of this work was to achieve a 7.4 VCF or greater at each of the three stages of the single pass diafiltration design to provide 99.75% buffer exchange. The study also aimed to maximize the feed concentration to minimize buffer requirements. An optimal feed concentration must be experimentally determined since the feed concentration is inversely proportional to the concentration factor that can be achieved at a given feed flux and pressure. In addition, conditions evaluated must not cause the product to over concentrate and plug the membrane.

Results from the DOE study are shown in Figure 4 and Table II. The model shows both feed concentration and flux are inversely correlated to the concentration factor. Pressure, however, appears to have no significant correlation in the range tested, as depicted in Figure 4. Pressure has been shown to impact the concentration factor under certain conditions (Casey et al, 2011), which suggests that the low feed flux, high protein concentration, and high viscosity of the buffer formulation used in this work resulted in operation in a pressure independent region of

the flux vs. TMP curves (Tutunjian, 1983). Based on the DOE study results shown in Table II, the optimal SPDF feed concentration was 15 g/L at a feed flux of 12 LMH. These conditions result in a VCF of 9.9, final product concentration of 148.5 g/L and diafiltration buffer use of 0.18 L<sub>buffer</sub>/ g<sub>protein</sub>. The DOE results in Figure 4 show that in theory a VCF of 7.4 can be achieved with a feed concentration as high as 19.8 g/L and a feed flux of 10.5 LMH. At these conditions, the buffer use would be 0.13 L<sub>buffer</sub>/ g<sub>protein</sub>, 1.5 times the buffer requirement of the recirculated diafiltration process performed at 70 g/L.

### **SPDF Clearance Study**

Antibody product was spiked with cupric sulfate and run through a 3-stage SPDF system to experimentally demonstrate that the required buffer exchange could be achieved. Cupric sulfate was selected as a positively charged process related impurity for which prior data showed acceptable clearance in a recirculated UFDF process (data not shown) and could be measured in the presence of the diafiltration buffer. The target small molecule clearance was 99.75% or 2.6 logs. The feed fluxes (10.5 LMH and 12 LMH) and feed concentration (16.6 g/L) used for this study were determined from the DOE study. Figure 5 shows the percent of cupric sulfate retained after dilution in each of the three retentate vessels. In addition to the measured cupric sulfate values, the blue color of the solution decreased in intensity across the stages. The final retentate was completely clear. At 10.5 LMH, the final stage retained copper at a concentration of 0.20% of the feed material which achieved the clearance target of greater than 99.75%. At 12 LMH, SPDF product had a final copper concentration of 0.27% of the initial, slightly above the target.

Although the target 99.75% excipient clearance was met during the experiment, the

measured copper concentration was higher than the theoretically calculated excipient amount for all points measured with one exception attributed to measurement variability. The higher measured copper concentrations may also be attributed to measurement variability.

Alternatively, they may be attributed to protein-solute interactions, rejection of the solute by the membrane (rejection coefficient is not equal to 1), or non-ideal flow within the cassettes. The Donnan Effect is not considered a likely cause for the difference due to positive charges of both the mAb and copper. The copper concentration on the retentate side of the membrane is expected to be lower than theoretically calculated based on the concentration factors to maintain electro neutrality (Bolton et al., 2011). A lower feed concentration can be used to achieve higher concentration factors and higher rates of buffer exchange, if a safety factor is required.

### **Continuous End-to-End Process**

A fully-continuous antibody cell culture and purification process, including diafiltration and final concentration, was performed. The pooled drug substance demonstrated equivalent or improved product quality when compared to the batch process (data not shown). The 3-stage SPDF system, followed by a single SPTFF membrane for final concentration, was operated for a six hour period without intervention. There was no significant change in feed pressure to suggest fouling. The retentate vessel levels remained constant due to the siphoned diafiltration buffer.

The feed and retentate concentrations of each of the three SPTFF membranes for diafiltration and single SPTFF membrane for final concentration were measured hourly during the continuous operation and are shown in Figure 6. Figure 6 also shows the VCF achieved across each SPTFF membrane as a function of time with the system operated at 12 LMH. As anticipated due to the difference in the buffer composition of the feed material, the VCFs across

the first membrane were lower than that of the remaining membranes. All variation in concentration factors for each membrane correlated with the feed concentration, with the exception of hour six of the first membrane. The sample taken at hour six was collected as the process was coming to an end and the error is likely due to the shutdown procedures. Studies that have operated the membranes for longer durations (96 hours) do not show progressive membrane fouling (data not shown). The calculated impurity clearance of the six hour end-to-end continuous process, based on the dilution factors, is shown in Table III. Diafiltration of 99.75% ( $\pm 0.15\%$ ) was achieved for the duration of the run.

## Discussion

This work demonstrates a viable single pass diafiltration design for use in a fully continuous mAb purification process. The design consists of three identical stages of concentration and dilution in series to achieve 99.75% buffer exchange, equivalent to a 6 diavolume traditional diafiltration step. SPDF was demonstrated over a six hour fully continuous run. Acceptable buffer exchange was demonstrated with a spiking study that measured excipient concentrations. In addition, a DOE study outlined the optimal feed concentration and flux to minimize buffer demand and showed theoretical buffer requirement can be 1.5 times that of the recirculated diafiltration process. The SPDF design proposed can be used to meet the buffer exchange associated with an eight or ten diavolume recirculated diafiltration process with the addition of a fourth or fifth stage, respectively, and the 7.4-fold concentration and dilution factor used in this work.

A safety factor will need to be employed when the system is operated for clinical or commercial applications to ensure the required buffer exchange is met, which will result in a slight increase in buffer use. A lower feed concentration than that determined to be optimal can

be used to account for a margin of error, while the feed flux and number of SPDF stages are held constant to minimize complexity. Alternatively, a future improvement to the system could incorporate inline measurement of concentration and feedback to adjust the feed flow rate in real-time. Inline measurement of concentration would also enable inline dilution and excipient spike for continuous formulation of the product.

Although the SPDF buffer demand is higher than that of a recirculated diafiltration process, the volume is expected to be acceptable based on buffer preparation and hold tank capacities of a typical manufacturing facility. In addition, buffer savings realized with other continuous purification technologies, such as multi column chromatography, can compensate for the slight increase in buffer used for the SPDF step. Countercurrent diafiltration methods can also be investigated to reduce the buffer demand (Nambiar, 2018).

The benefits of SPDF are most readily realized when run as part of a fully continuous mAb purification process. In a fully continuous mAb process, the diafiltration step is performed consecutively with prior purification steps and greatly reduces the overall process time. Use of traditional recirculated diafiltration after a continuous purification process would require greater process time for the product to be pooled prior to diafiltration.

SPDF also demonstrates benefits for traditional batch processes. The SPDF design removes limitations on high and low product concentrations due to available UFDF tank sizes. A recirculated system must be large enough to hold the entire product volume during the diafiltration step and have a low enough minimum hold up volume to accommodate the concentrated product. In many instances, these two requirements can exceed physical and technical limitations of a single recirculated TFF system. For SPDF, the vessel volume is



limited only by the need to mix the diafiltration buffer with the high concentration retentate and maintain solution homogeneity.

Another benefit of the SPDF design is the option to use significantly smaller piping and tanks when compared to a traditional recirculated diafiltration system, as shown in Table IV for 200, 2k and 20k liter bioreactor scales. The SPDF flow rates, which are orders of magnitude lower than those for recirculated UFDF, enable the use of disposable tubing even at the 20,000 L bioreactor scale. Despite three vessels in the SPDF system, the SPDF retentate vessel's total volume is two orders of magnitude smaller than the recirculated UFDF tanks. As a result of reduced tank sizes, facilities require less footprint, require less capital investment and can switch to single use vessels at all scales evaluated.

SPDF may also be the preferred diafiltration method for products sensitive to aggregation due to the reduced number of pump passes and limited residence time in stirred retentate vessels (Callahan et al., 2014). Since SPDF does not recirculate product, products can experience up to 10 times fewer pump passes compared to recirculated diafiltration systems.

Although there are many opportunities for the SPDF technology, some challenges do exist. Diafiltration is the rate controlling step of a continuous purification process since feed flux and feed concentration control the concentration factor for a given SPTFF membrane design. SPTFF membranes are only available in a limited number of sizes and therefore limit the volumetric and mass flow rate of the entire continuous purification train. However, as the technology gains acceptance, a greater number of size options, both large and small, is not unforeseeable and would facilitate more widespread use of the devices. In addition, future work will need to be performed to demonstrate that the system can be cleaned, sanitized and operated in a way to maintain bioburden control for extended durations. Although these challenges do

exist, they are not expected to limit implementation in clinical or commercial settings as continuous manufacturing approaches are more widely adopted.

## **Conclusions**

This work has demonstrated the feasibility of single pass diafiltration in an end-to-end continuous mAb purification process. Buffer exchange of 99.75% with SPDF, equivalent to a six diavolume recirculated diafiltration process, was demonstrated with a monoclonal antibody product. The SPDF system included several design factors to minimize equipment complexity and maximize buffer efficiency. The work included a spiking study to model small molecule clearance and extended operation during a continuous end-to-end mAb purification process was demonstrated. SPDF is expected to become a staple of continuous mAb purification processes in the biopharmaceutical industry.

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## **Conflicts of Interest**

The authors have no conflicts of interest to declare.

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Table I. Continuous End-to-End Process Materials & Equipment.

Table II. Experimental conditions and results for DOE study to identify process conditions that achieve 99.75% buffer exchange with a 3-stage diafiltration process.

Table III. Target impurity clearance achieved for each hour of a mAb purification run.

Table IV. Comparison of recirculated and single pass UFDF equipment size.

Table I. Continuous End-to-End Process Materials & Equipment.

| Step # | Step Description                                            | Materials & Equipment                                                                                                                                                                                                                                                                 |
|--------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Perfusion Cell Culture                                      | 50L Bioreactor from HyClone Laboratories with a Valley Instruments Controller                                                                                                                                                                                                         |
| 2      | Perfusion Harvest                                           | Repligen XCell ATF 6 Single-Use System                                                                                                                                                                                                                                                |
| 3      | Affinity Chromatography (Protein A)                         | NovaSep BioSC MultiColumn-Chromatography System, five Millipore Vantage L Chromatography Columns with 2.2 cm diameters packed to a 5cm bed height<br>MabSelect SuRe Protein A Resin from GE Healthcare                                                                                |
| 4      | Low pH Viral Inactivation and Neutralization                | Koflo Unmounted In-Line Mixer from Cole Parmer<br>XK 50/30 GE Column<br>Superdex 200 Size Exclusion Resin                                                                                                                                                                             |
| 5      | Filter Train (Includes Anion Exchange and Viral Filtration) | 1. Millipore Millistak+® X0HC Pod Depth Filter<br>2. Millipore SPTFF Element with two 88 cm <sup>2</sup> membranes in series<br>3. Millipore Express® SHC 0.5/0.2 µm Hydrophilic Filter<br>4. Pall Mustang Q Anion Exchange Membrane<br>5. Planova™ 20N Viral Filter from Asahi Kasei |
| 6      | Diafiltration (SPDF) & Concentration (SPTFF)                | Four Pall Corporation Cadence Inline Concentrator (ILC) 4-in-series modules, 30 kD molecular weight cutoff, Delta regenerated cellulose, 0.065 m <sup>2</sup> . Modules are designed with a step-wise decrease in membrane cross sectional area.                                      |

Table II. Experimental conditions and results for DOE study to identify process conditions that achieve 99.75% buffer exchange with a 3-stage diafiltration process.

| <b>Feed Concentration (mg/mL)</b> | <b>Feed Flux (LMH<sup>1</sup>)</b> | <b>Feed Pressure (psi)</b> | <b>Volumetric Concentration Factor</b> |
|-----------------------------------|------------------------------------|----------------------------|----------------------------------------|
| 25                                | 13.5                               | 25                         | 3.8                                    |
| 25                                | 10.5                               | 25                         | 5.2                                    |
| 25                                | 13.5                               | 15                         | 4.0                                    |
| 25                                | 12                                 | 20                         | 4.5                                    |
| 25                                | 10.5                               | 15                         | 5.5                                    |
| 15                                | 12                                 | 20                         | 9.9 *                                  |
| 15                                | 13.5                               | 15                         | 6.9                                    |
| 15                                | 10.5                               | 25                         | 11.4 *                                 |
| 15                                | 13.5                               | 25                         | 6.9                                    |
| 15                                | 10.5                               | 15                         | 11.2 *                                 |
| 20                                | 12                                 | 15                         | 5.5                                    |
| 20                                | 12                                 | 20                         | 6.1                                    |
| 20                                | 13.5                               | 20                         | 5.4                                    |
| 20                                | 12                                 | 20                         | 6.1                                    |
| 20                                | 10.5                               | 20                         | 6.7                                    |
| 20                                | 12                                 | 25                         | 6.1                                    |

<sup>1</sup>LMH = L/m<sup>2</sup>/hr

\* Acceptable volumetric concentration factor (i.e. at or above 7.4).

Table III. Target impurity clearance achieved for each hour of a mAb purification run.

| Time Point (hrs) | Buffer Exchange (%)* |
|------------------|----------------------|
| Hour 1           | 99.76                |
| Hour 2           | 99.75                |
| Hour 3           | 99.83                |
| Hour 4           | 99.77                |
| Hour 5           | 99.60                |
| Hour 6           | 99.70                |

\* Target impurity clearance was 99.75%.



Table IV. Comparison of recirculated and single pass UFDF equipment size.

|                                 | <b>Recirculated<br/>200 L Scale</b> | <b>Single Pass<br/>200 L Scale</b>              | <b>Single Pass<br/>2,000 L Scale</b>         | <b>Single Pass<br/>20,000 L Scale</b>             |
|---------------------------------|-------------------------------------|-------------------------------------------------|----------------------------------------------|---------------------------------------------------|
| UFDF Membrane Area <sup>1</sup> | 1 m <sup>2</sup>                    | 0.26 m <sup>2</sup> (4 x 0.065 m <sup>2</sup> ) | 2.8 m <sup>2</sup> (4 x 0.7 m <sup>2</sup> ) | 28.0 m <sup>2</sup> (4 x 2 x 3.5 m <sup>2</sup> ) |
| Flow Rate <sup>1</sup>          | 6 L/min                             | 13.0 ml/min                                     | 140 ml/min                                   | 1.4 L/min                                         |
| Piping Diameter <sup>2</sup>    | 0.5 in                              | 1/32 in ID                                      | 1/8 in ID                                    | 1/4 in ID                                         |
| Vessel Volume <sup>3</sup>      | 14 L                                | 130 ml                                          | 1.4 L                                        | 14 L                                              |

<sup>1</sup> Assumes 5 g/L titer, 100% yield, 5 day run rate, 0.2 L/min/m<sup>2</sup> through SPDF step or 1000 g/m<sup>2</sup> for recirculated UFDF.

<sup>2</sup> Assumes 2.5 ft/sec max flow to minimize pressure.

<sup>3</sup> Assumes residence time of 10 min for SPDF, vessel volume to hold all product at 70 g/L for recirculated UFDF.

Figure 1. A 3-stage diafiltration process equipment setup.

Figure 2. Buffer requirements for 99.75% buffer exchange.

Figure 3. Buffer efficiency of diafiltration process improves with increased mAb concentration at SPTFF retentate, prior to dilution.

Figure 4. Illustration of operating space (white) that provides  $\geq 7.5$  concentration factor to achieve 99.75% buffer exchange with a 3-stage diafiltration process (top). Modeled effects of feed concentration, feed flux and feed pressure on concentration factor (bottom).

Figure 5. Cupric sulfate concentration after each of three diafiltration stages, as a percentage of the original spiked concentration.

Figure 6. Concentration profile, feed concentrations and concentration factors of a 3-stage diafiltration process during each hour of a continuous mAb purification run.

Figure 1

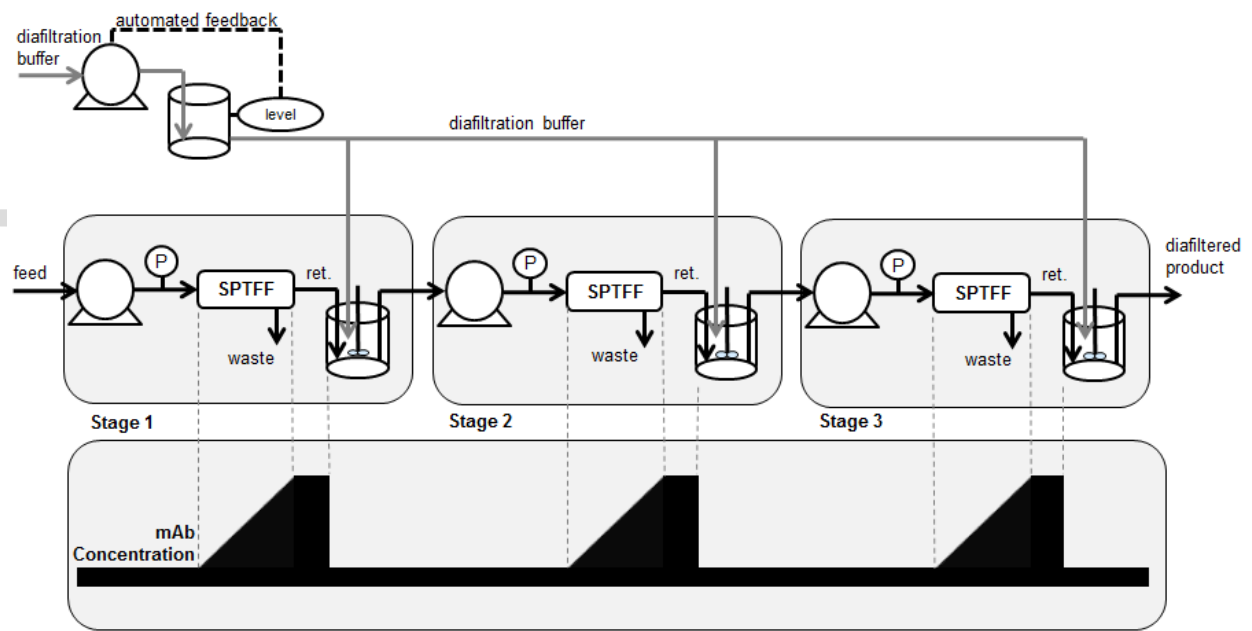


Figure 2

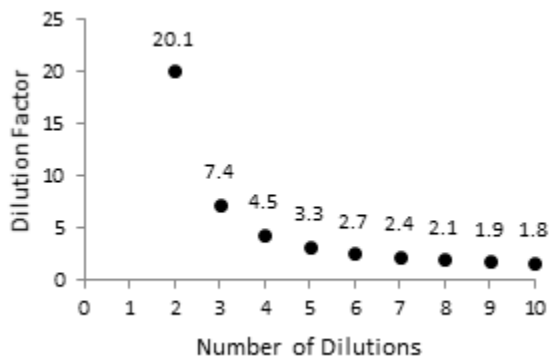


Figure 3

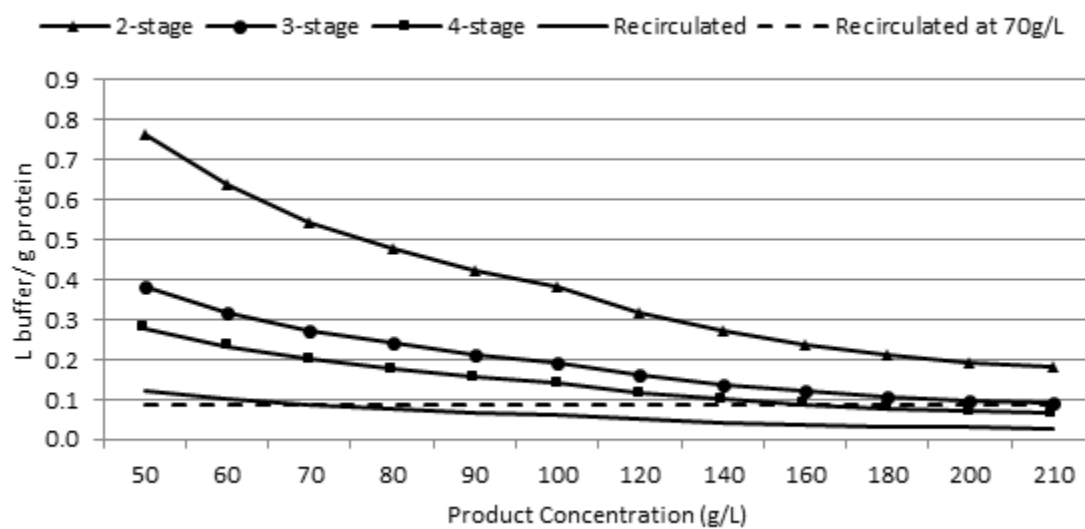


Figure 4

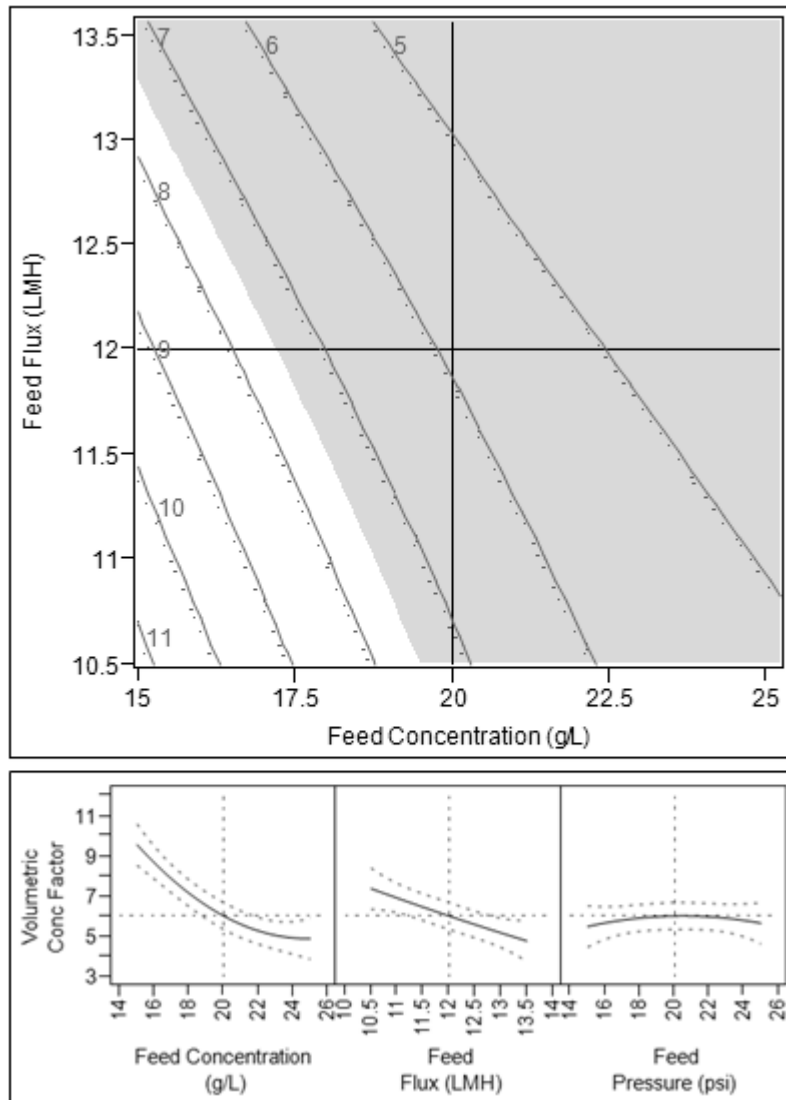


Figure 5

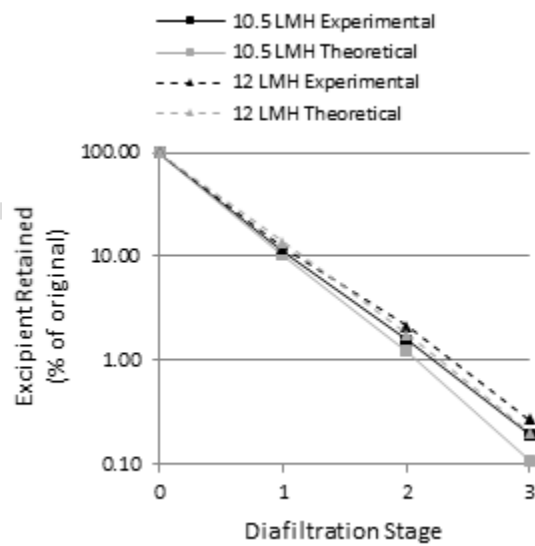


Figure 6

